

Nitrogen Management: Opportunities for Taiwan's Net-Zero Transition

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KEY CLAIM

Agriculture is central to Taiwan's nitrogen challenge—and sustainable nitrogen management offers important opportunities for climate mitigation and the net-zero transition

BRIEF JUSTIFICATION, REMAINING CHALLENGES

Agriculture is a major source of reactive nitrogen emissions, primarily through fertilizer use and livestock production. Taiwan's 20-year nitrogen footprint trends can reveal how agricultural production and consumption patterns have changed over time. Improving nitrogen use efficiency and reducing nitrogen losses can therefore deliver **multiple benefits**, including lower environmental pollution, improved resource efficiency, and reduced nitrous oxide (N₂O) emissions—supporting Taiwan's net-zero transition.

The remaining challenges are listed as follows:

- Nitrogen is not yet fully integrated into net-zero and climate policies.
- Agricultural nitrogen losses remain difficult to reduce, particularly from fertilizer and livestock systems.
- Trade-offs between food production, food security, and environmental protection need to be addressed.
- A lack of integrated indicators and long-term monitoring makes it difficult to evaluate progress across the food and agricultural system.

POTENTIAL FOR SYNERGY WITH OTHER CONTRIBUTIONS

Sustainable nitrogen management has strong potential to reduce agricultural N losses and N₂O emissions while improving nitrogen-use efficiency, food security, water and air quality, biodiversity, and progress toward net zero.

QUESTIONS TO SEED THE DISCUSSION

1. How can nitrogen management support both food security and net-zero goals?
2. What policy changes are needed to turn this potential into action?

REFERENCES

References should follow a simple author–date style, for example:

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